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भारतीय प्रौद्योगिकी संस्थान मुंबई
INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

उत्तर पुस्तिका/ Answer Book-16

22IBC163
रोल नं./**Roll No.**

CE313
पाठ्यक्रम नाम/**Course Name**

शाखा/प्रभाग/**Branch/Div.**शैक्षणिक बैच /**Tutorial Batch**

अनुभाग/**Section**

प्र.सं/Q. No.	1	2	3	4	5	6	7	8	9	10	11	12	योग/Total
आरंभिक पृष्ठ Starting Page No.	01	13	13	17									43
अंक/Marks													

प्र.सं/Q. No.	13	14	15	16	17	18	19	20	21	22	23	24	योग/Total
आरंभिक पृष्ठ Starting Page No.													
अंक/Marks													

परीक्षा में उपस्थित होने वाले अभ्यर्थियों से घोषणा पत्र	**Declaration from the Candidate appearing in the Examination**
1. मैं, इस परीक्षा के आयोजन संबंधी पाठ्यक्रम अनुदेशक द्वारा दिए गए सभी अनुदेशों का पालन करूँगा / करूँगी।	1. I will abide by all instructions given by the course instructor regarding the conduct of this examination.
2. मैं, संस्थान के नियम तथा विनियमों में दिए गए शैक्षिक कदाचार संबंधी नियम तथा विनियमों से अवगत हूँ।	2. I am aware of the rules and regulations regarding academic misconduct as given in rules and regulations of the institute.
3. मैं, इस परीक्षा के दौरान शैक्षिक सत्यनिष्ठा के सिद्धांतों पर दृढ़ रहूँगा / रहूँगी।	3. I will adhere to the principles of academic integrity during this examination.

हस्ताक्षर/ Signature :- Aadishree

दिनांक/ Date :- 24/2/25.

(3)

For the given data: $\bar{x} = 64.886$ ✓

$s = 17.735$ ✓

$$\alpha = \frac{\sqrt{6}}{\pi} s = 13.828$$ ✓

$$u = \bar{x} - 0.57721 \alpha$$

$$= 56.904$$ ✓

$$x_T = u + \alpha y_T$$ ✓

$$= 56.904 + 13.828\, y_T$$

$$y_T = -\ln\left(\ln\left(\frac{T}{T-1}\right)\right)$$ [T is in years].

Sr. No.	T (yrs)	y_T (mm)	x_T (mm)
1.	5	1.499	77.645
2.	10	2.250	88.017
3.	50	3.902	110.861
4.	100	4.600	120.513
5.	1000	6.907	152.414

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Show 1 set of calculations.

4. $n = 0.016$

$$i = \frac{3048\, T^{0.180}}{(T_d + 23)} = \frac{4072.204}{T_d + 23} \qquad [\because T = 5 \text{ yrs}]$$

$$\Rightarrow i = \frac{4072.204}{T_d + 23} \qquad (T_d \text{ is in minutes})$$

$$D = \left(\frac{n Q_p}{0.312\, s^{1/2}}\right)^{3/8}$$

Pipe - EB.:

Recieves flow only from catchment (III).

$C = 0.6$ $\quad A = 16187\ m^2$. $\quad \sum CA = 9712.2\ m^2$

$t_c = t_i = 10$ $\quad i = 123.4\ mm/hr = 0.0343 \times 10^{-3}\ m/sec$

$Q_p = CiA = 0.338\ m^3/sec.$

$$D = \left(\frac{n Q_p}{0.312\, s^{1/2}}\right)^{3/8} = 560.396\ mm.$$

$D_{av} = 600\ mm.$

$$v_f = \frac{Q_p}{(\pi D_{av}^2/4)} = 1.178\ m/s.$$

$$t_f = l/v_f = 116.324\ sec$$

$$t_f = 1.939\ min.$$

Pipe - AB:

Flow from (I) and (II).

$A = 8094 + 12141 = 20235\ m^2.$

$\sum CA = 0.7 \times 8094 + 0.7 \times 12141.$

$= 14164.5\ m^2$

$t_c = \max(t_1, t_2) = 7\ min.$

$i = 135.74\ mm/hr = 0.0377 \times 10^{-3}\ m/sec$

$Q_p = (\sum CA) \times i = 0.534\ m^3/sec.$

$D = 640.065\ mm.$ $\quad v_f = 1.388\ m/s$

$D_{av} = 700\ mm$ $\quad t_f = l/v_f = 121\ sec = 2.018\ min$

Pipe - BC :

Flow from (I), (II), (III), (IV), (V).

$A = 72843\ m^2$

$\Sigma CA = 43705.9\ m^2$

$t_c = \max(t_1, t_2, t_3, t_4, t_5)$

$= \max(5+2.018, 7+2.018, 10+1.939, 10, 15)$

$= 15.\ min$

$i = 107.163\ mm/hr = 0.0298 \times 10^{-3}\ m/sec.$

$Q_p = (\Sigma CA)\, i = 1.301\ m^3/sec.$

$D = 934\ mm$ $\qquad v_f = 1.655\ m/s.$

$D_{av} = 1000\ mm.$ $\qquad t_f = 73.707\ sec = 1.228\ min.$

Pipe - CD :

Flow from all catchments.

$A = 109265\ m^2.$

$\Sigma CA = 61916.9\ m^2.$

$t_c = \max(t)$

$= 15 + 1.228$ (from catchment (V))

$= 16.228\ min.$

$i = 103.809\ mm/hr = 0.0288 \times 10^{-3}\ m/sec.$

$Q_p = (\Sigma(CA)) \times i. = 1.785\ m^3/sec$

$D = 1144.231\ mm.$

$D_{av} = 1200\ mm.$

$v_f = 1.578\ m/sec$

$t_f = 86.8\ sec = 1.447\ min.$

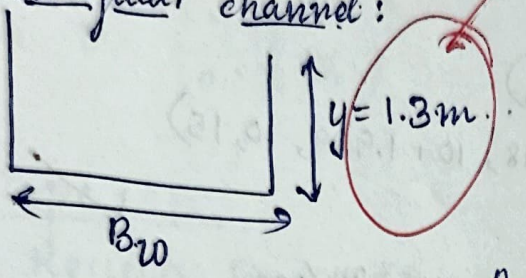
① .

$Q = 311\ m^3/sec.$ $\quad v = \frac{1}{n} R^{2/3} S_0^{1/2}$

$n = 0.016.$

$S_0 = 0.0039.$

Rectangular channel :



$A = B_W \times y.$

$= 1.3\ B_W$

$P = B_W + 2y = B_W + 2.6$

$R = \frac{A}{P} = \frac{1.3 B_W}{2.6 + B_W}$

$Q = Av.$

$\Rightarrow 311 = 1.3.\ B_W \times \frac{1}{n} \left(\frac{1.3 B_W}{2.6 + B_W} \right)^{2/3} \times 0.0039^{1/2}$

$\Rightarrow 51.457 = \frac{B_W^{5/3}}{(2.6 + B_W)^{2/3}}$

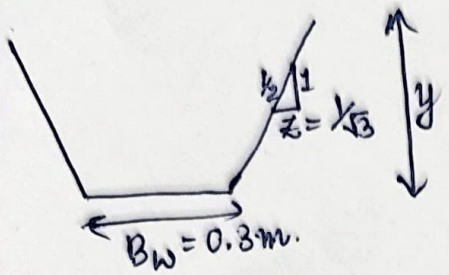
$\Rightarrow 186247.155 = \frac{B_W^5}{(2.6 + B_W)^2}$

$921030.77 + 708485.2\ B_W + B_W^2 (136247.155) = B_W^5.$

Trapezoidal channel :

$B_w = 0.3\,m$.

$z = 1/\sqrt{3}$.



$y = \frac{\sqrt{3}}{2} B_w =$

$B_w = \frac{2}{\sqrt{3}} y$

$A = (B_w + zy)y = \left(\frac{2}{\sqrt{3}}y + \frac{1}{\sqrt{3}}y\right)y = \sqrt{3}\,y^2$

$P = 3 \times B_w = 2\sqrt{3}\,y$.

$R = A/P = y/2$.

$Q = A \times \frac{1}{n} R^{2/3} S^{1/2} = \frac{\sqrt{3}\,y^2 (y/2)^{2/3}}{n} \times \sqrt{0.0039}$

$\Rightarrow y^{8/3} = 73.025$.

$\Rightarrow y = 4.998\,m$

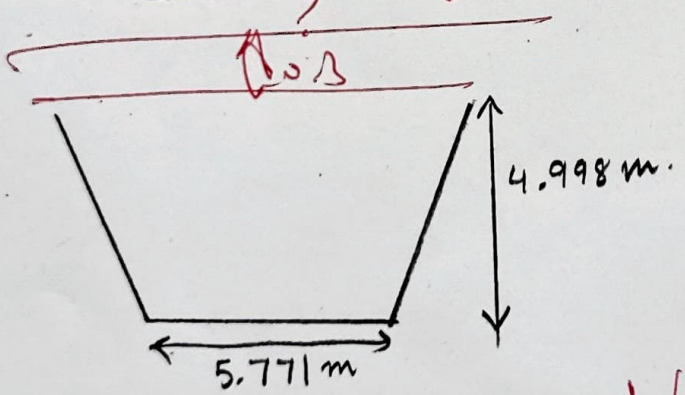
$B_w = 5.771\,m$.

$v = 7.187\,m/s$.

$S_0 = 0.0039$

freeboard = 0.3m.

$B_t = 11.542\,m$.



V = ? Safe or not

07

PLEASE WRITE YOUR ROLL NUMBER **Roll No.** 22B0763

CODE **A**		CE 343 **S2** Water Resources Engineering 24 February 2024 Monday 1130-1330 (SLOT 3) Time: 2 h **weightage: 30 %**

Please write your roll number on the question paper and the rainfall sheet. Attempt all FOUR questions; Write the question number clearly and begin answer to each question on a new page. Answers should be computed step by step and to 3 decimal places. Make suitable assumptions where necessary. There is no partial marking!!! Take Manning's roughness coefficient n = 0.016 wherever required. No attachments, only single answerbook!

Available pipe diameters (mm) are:
100,150,200,225,250,300,350,400,450,500,600,700,800,900,1000,1100,1200,1400,1600,1800,2000 and 2200.

1 Design the **best** hydraulic ***trapezoidal*** section for a non-erodible channel to carry a peak discharge of 311 m^3/s. Take Manning's n =0.016 and S_0= 0.0039. Provide a freeboard of 0.3 m. *Due to site constraints, the maximum available depth is **1.3** m and a **rectangular** channel is desired. What would then be the channel section? Check for self-cleansing velocity.* Give a comparative table with a clear sketch of the final cross-sections with dimensions and table showing depth, freeboard, bottom width, top width, slope, velocity of both sections. **(15)**

2 The rainfall during the short intense event of 4th July 2015 over the Civil Engg Dept at IIT Bombay is given in the attached table. Carry out the computations of rainfall depth and intensity for the durations of 1, 3, 5, 10 and 15 minutes on the sheet. What are corresponding maximum depths and maximum intensities? Plot the intensity duration curve on the given graph paper (Write your roll no. as well!) **(15)**

3 The annual maximum values of 15 - minute duration rainfall (in mm) from an AWS station from 1973 to 2010 are shown below. Carry out a storm frequency analysis using Extreme Value Type I distribution and calculate the 5-, 10-, 50-, 100- and 1000-year return period maximum values of 15-minute rainfall at the AWS station. **(15)**

Year	1970	1980	1990	2000
0		53	33	34
1		76	96	70
2		57	94	57
3	49	80	80	92
4	66	66	62	66
5	36	68	71	65
6	58	68	111	63
7	41	61	64	60
8	47	88	52	
9	74	49	64	

PLEASE WRITE YOUR ROLL NUMBER **Roll No.** 22B5763

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(15)

A simplified drainage basin consisting of 7 catchments is shown in Figure 1. The area, runoff coefficients and inlet time for each sub-catchment are shown in the Table 1 while the length and slope for each pipe are shown in Table 2. Assume Manning's n = **0.016**. The design rainfall intensity for this catchment is given by

$$i = 3048\ T^{0.180} / (T_d + 23)$$

where i is the rainfall intensity in mm/h; **T** is the return period in years and T_d is the duration in minutes.

Determine the diameters for the pipes AB, BC, CD and EB for a return period of five years. All calculation steps **MUST** be clearly shown in the answerbook. **Summarise your results in Table 2 and attach it with your answerbook. IS 458:2021** specifies concrete pipes of class NP3 diameters(mm):80,100,150,200,225,250,300,350,400,450,500,600,700, 800, 900,1000,1100,1200,1400,1600,1800,2000,2200,2400,2600. (If necessary, assume suitable data and provide proper justification for the assumption)

Figure 1

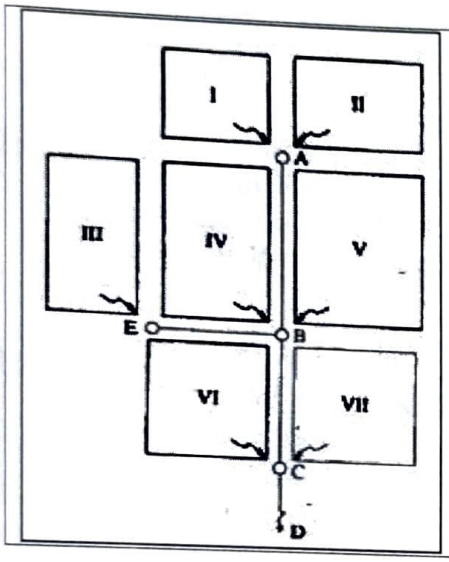


Table 1

S No	Catchment	Area (m^2)	C	Inlet time t_i (min)
1	I.	8094	0.7	5
2	II.	12141	0.7	7
3	III.	16187	0.6	10
4	IV.	16187	0.6	10
5	V.	20234	0.5	15
6	VI.	18211	0.5	15
7	VII.	18211	0.5	15

Table 2

Pipe	Length L (m)	Slope S_o m/m	Total area drained A (m^2)	ΣCA	Time of Conc. t_c (min)	Rainfall intensity I (mm/hr)	Design Dis-charge Q (m^3/s)	Computed dia (mm)	Dia. Avail able (mm)	Flow vel. (m/s)	Flow time (min)
EB	137	0.0064	16187	9712.2	10	123.400	0.333	560.396	600	1.178	1.939
AB	168	0.0081	20235	14164.5	7	135.74	0.534	640.065	700	1.388	2.018
BC	122	0.0064	72843	43705.9	15	107.163	1.301	934.184	1000	1.655	1.228
CD	137	0.0064	109265	61916.9	16.228	103.809	1.785	1144.231	1200	1.578	1.447

---------- END ----------

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0768
(15)

Roll No. 22BC768

	Observed data				Running totals			
Date	Time	Time	Rain (mm)	Cumulative rainfall	3 min	5 min	10 min	15 min
04-07-2015	7:00 PM	0	0	0				
04-07-2015	7:01 PM	1	0.76	0.76				
04-07-2015	7:02 PM	2	0.76	1.52				
04-07-2015	7:03 PM	3	1.02	2.54	2.54			
04-07-2015	7:04 PM	4	0.51	3.05	2.29			
04-07-2015	7:05 PM	5	0.51	3.56	2.04	3.56		
04-07-2015	7:06 PM	6	0.51	4.07	1.53	3.31		
04-07-2015	7:07 PM	7	0.25	4.32	1.27	2.8		
04-07-2015	7:08 PM	8	0.51	4.83	1.27	2.29		
04-07-2015	7:09 PM	9	0.25	5.08	1.01	2.03		
04-07-2015	7:10 PM	10	0.25	5.33	1.01	1.77	5.33	
04-07-2015	7:11 PM	11	0.51	5.84	1.01	1.77	5.08	
04-07-2015	7:12 PM	12	0.25	6.09	1.01	1.77	4.57	
04-07-2015	7:13 PM	13	0.51	6.6	1.27	1.77	4.06	
04-07-2015	7:14 PM	14	0.25	6.85	1.01	1.77	3.8	
04-07-2015	7:15 PM	15	0.25	7.1	1.01	1.77	3.54	7.1
04-07-2015	7:16 PM	16	0.25	7.35	0.75	1.51	3.28	6.59
04-07-2015	7:17 PM	17	0.25	7.6	0.75	1.51	3.28	6.08
04-07-2015	7:18 PM	18	0.51	8.11	1.01	1.51	3.28	5.57
04-07-2015	7:19 PM	19	0.25	8.36	1.01	1.51	3.28	5.31
04-07-2015	7:20 PM	20	0.51	8.87	1.27	1.77	3.54	5.31
04-07-2015	7:21 PM	21	0.51	9.38	1.27	2.03	3.54	5.31
04-07-2015	7:22 PM	22	0.76	10.14	1.78	2.54	4.05	5.82
04-07-2015	7:23 PM	23	0.51	10.65	1.78	2.54	4.05	5.82
04-07-2015	7:24 PM	24	0.51	11.16	1.78	2.8	4.31	6.08
04-07-2015	7:25 PM	25	0.25	11.41	1.27	2.54	4.31	6.08
04-07-2015	7:26 PM	26	0.76	12.17	1.52	2.79	5.36	6.33
04-07-2015	7:27 PM	27	1.02	13.19	2.03	3.05	5.59	7.1
04-07-2015	7:28 PM	28	0.51	13.7	2.29	3.05	5.59	7.1
04-07-2015	7:29 PM	29	0.51	14.21	2.04	3.05	5.85	7.36
04-07-2015	7:30 PM	30	0.25	14.46	1.27	3.05	5.59	7.36
Max depth (mm)					2.54	3.56	5.85	7.36
Rainfall Int (mm/h)					50.8	42.72	35.1	29.44

?

Missing

Need Qn carefully

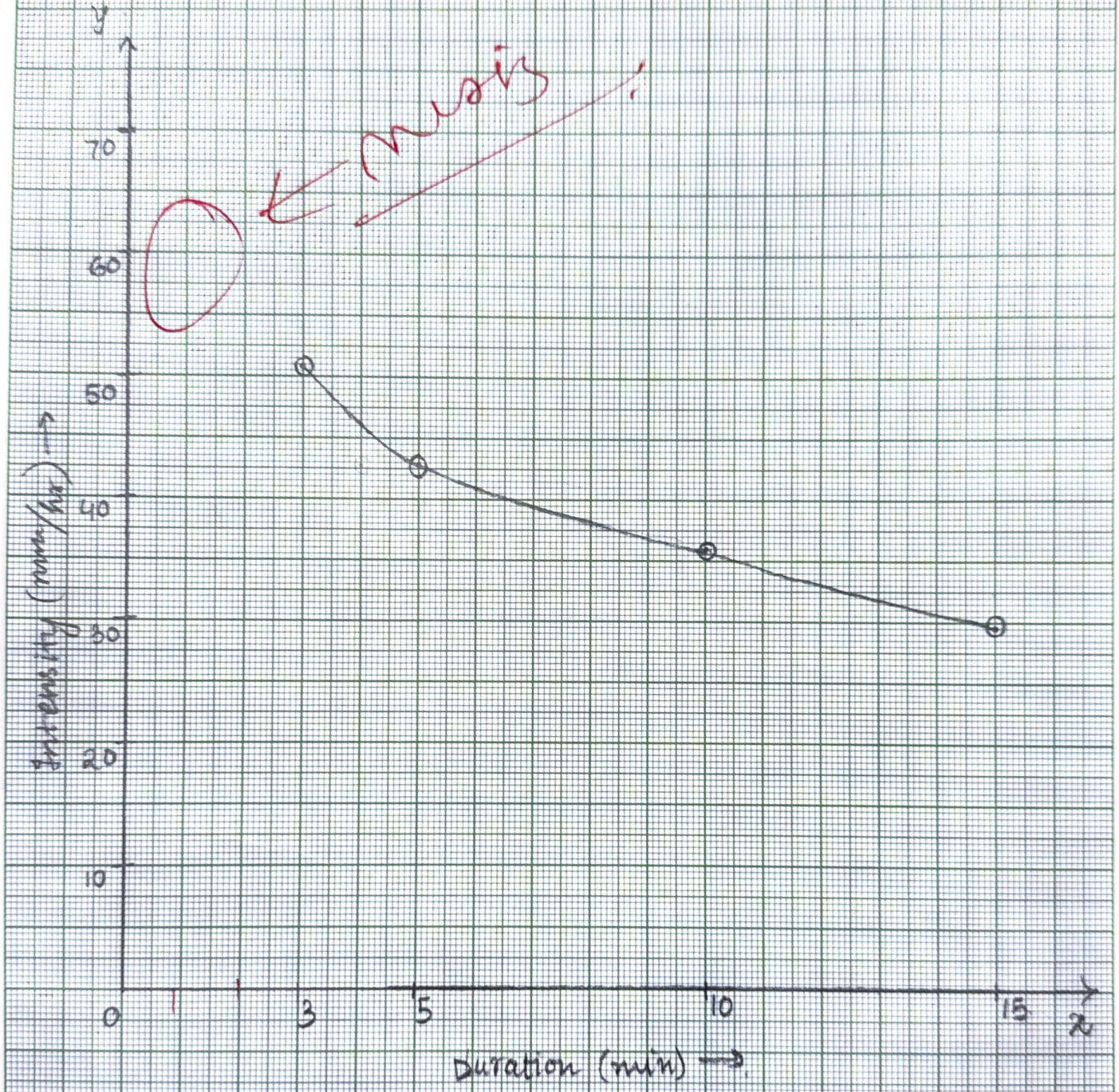
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(2)

Scale

y-axis: 1 cm = 5 mm/hr

x-axis: 1 cm = 1 min.



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